# Professional Software Engineering Processes and Human Factors 811373A: Scrum project

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## 1 Project overview and objectives

In this project, the groups will develop a PQC Security Monitoring Dashboard for an organization that is migrating its systems to use ML-KEM-based post-quantum cryptography. The dashboard should help the stakeholders of the organization. The main purpose of the project is NOT to build an advanced cybersecurity tool. Instead, the project is designed to help you practice professional software engineering processes and human factors related to them. The dashboard acts as a realistic project case through which you will learn and apply Scrum. The dashboard should provide meaningful information about PQC, such as Harvest-Now-Decrypt-Later (HNDL), downgrade risks, and security status. Synthetic data should be used. The dashboard must be generated using an LLM. The LLM-generated content must be evaluated. However, the focus is on the project work processes and stakeholder-oriented views, rather than implementing complex monitoring technologies.

The project will be conducted using Scrum and Jira, divided into three sprints. Also, the student groups will create and maintain a product backlog, define and prioritize user stories, estimate work, track progress in Jira, and participate in sprint planning, sprint reviews, and sprint retrospectives.

There will be three (3) sprints. Each sprint must produce a working dashboard increment. The teacher of the course will act as the product owner. During the project, requirements may change, priorities may shift, and simulated events, such as a security incident, may be introduced. The groups are expected to adapt and respond to these changes using Scrum practices.

Assessment focuses primarily on the professional software engineering process and human factors: 60% of the grade is based on project management, Scrum practices, Jira usage, stakeholder and requirements management, groupwork, communication, risk management, adaptation to change, and reflection on human factors. In addition, 20% is based on the dashboard prototype and its ability to communicate relevant information clearly and effectively including the evaluation of the LLM-generated code. The remaining 20% comes from the group presentation and individual reflection, 10 % each.

## 2 How to complete the work

- **Groups:** Form groups of five students. The project cannot be completed individually.
- **Group setup:** Agree on how your group communicates, makes decisions, and handles disagreements. Assign Scrum roles; the Scrum Master role should rotate between sprints.
- **Product vision and stakeholders:** Write a short product vision and a stakeholder analysis describing who the dashboard serves, what each stakeholder needs, and where their needs conflict.
- **Product backlog in Jira:** Write user stories with acceptance criteria, estimate and prioritize them, and keep the backlog up to date.
- **Risks and decisions:** Maintain a risk register and a decision log throughout the project.
- **Responding to change:** When requirements change or a simulated event occurs, analyze its impact, re-prioritize with the product owner, and document what you changed and why.
- **Dashboard:** Build the dashboard incrementally using synthetic data. Be explicit about what each indicator can and cannot tell. In the end of every sprint, there must be a functioning version of the dashboard.
- **Sprints:** Complete three sprints. For each sprint, set a sprint goal in sprint planning, track progress in Jira, demonstrate a working increment in the sprint review, and hold a retrospective. Record retrospective actions in Jira and follow them up in the next sprint.
- **Sprint reports:** Each sprint submission includes a 5-minute video presentation, the presentation slides used in the video as a PDF file, and a written sprint report of 500 words, which the video presentation and slides are based on.

  In the report (video, slides, text), describe the sprint goal and outcome, planned versus completed work, key decisions and their rationale, changes in risks, and retrospective actions, and evaluate the content produced by LLM.

  In the reports, walk through the development of your Jira project and the incremental development of the dashboard during the sprint. Also, refer to the product vision, stakeholders, risks, decisions, and responses to change in the sprint reports.

## 3 Submissions and grading

- Sprint reports: total of 60 % (20 % each)
- Dashboard: 20 %
- Final presentation: 10 %
- Individual reflection 10 %

## 4 Schedule

- Project starts: Mon 2nd Nov
- DL for forming the groups: Thu 5th Nov
- Project final DL: Sun 20th Dec
- DLs for the sprints:
  1. Wed 11th Nov
  2. Wed 18th Nov
  3. Wed 2nd Dec
- DL for the individual reflection: Sun 20th Dec
- DL for the substitute assignment: Sun 20th Dec
- Final group presentations: Tue and Wed, 15th and 16th Dec
- Project sessions:
  - Thu 12.11. (lecture hall and online)
  - Thu 19.11. (lecture hall and online)
  - Thu 26.11. (online only)
- Final DL and the course ends: Sun 20th Dec
- All DL times are at 23:59 Finnish time

## 5 Contact

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